

Configuring VRRP to Implement Gateway Redundancy

Difficulty (out of 3 stars): ★★★

Recommended study time: 2 hours

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About This Lab

Task Description

Virtual Router Redundancy Protocol (VRRP) is a fault tolerance protocol. It elects the master router from multiple routers according to specific rules. The master router forwards data packets and uses its virtual MAC address as the ARP source address. Other routers function as backup routers. The master router periodically sends VRRP multicast packets to the backup routers, ensuring that the backup routers can learn the master router's status in a timely manner. In normal cases, backup routers do not forward packets destined for the virtual MAC address. If the master router fails and backup routers do not receive any VRRP packets within a specified period, a backup router takes over services from the master router, ensuring communication continuity and reliability.

After completing this task, you will have a clear understanding of VRRP and be familiar with how to use VRRP to construct and configure gateway redundancy.

Objectives

On completing this task, you will be able to:

1. Deploy VRRP.
2. Implement collaboration between VRRP and MSTP.
3. Configure association between BFD and VRRP.

Task Preparation

Network Topology

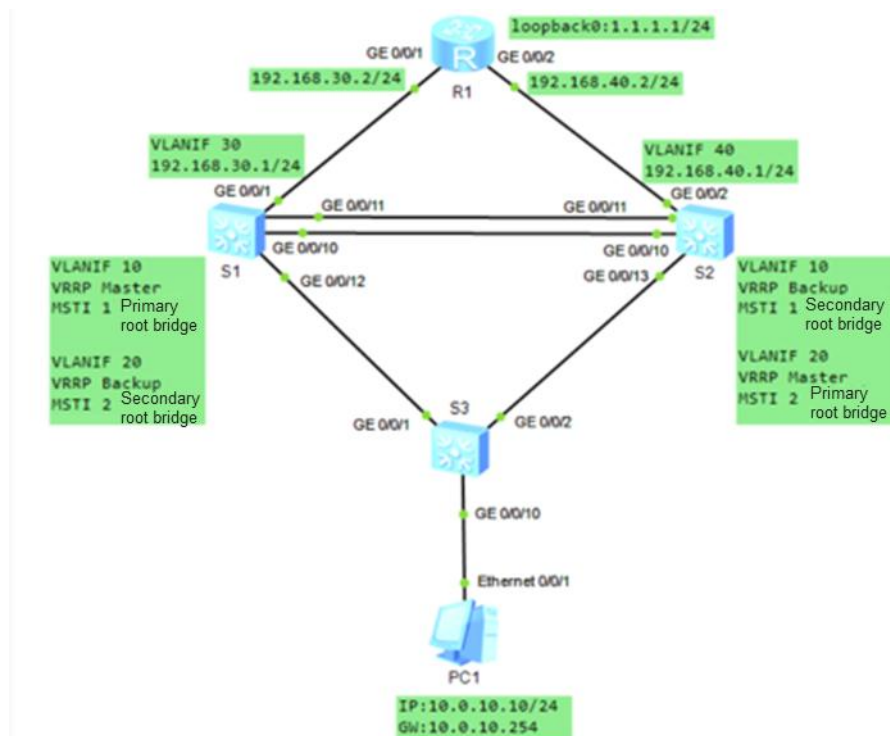


Figure 1 Topology for configuring VRRP to implement gateway redundancy

1. Devices are connected as shown in the preceding figure. VLAN 10 and VLAN 20 exist on the network, each with a VRRP group configured. The IDs of the VLANs are used as the VRIDs for their respective VRRP groups. S1 is configured as the master of the VRRP group in VLAN 10, and S2 as the master of the VRRP group in VLAN 20.
2. In addition, MSTP is deployed on S1, S2, and S3, and instances 1 and 2 are created. VLAN 10 is mapped to MSTI 1, and VLAN 20 is mapped to MSTI 2. S1 is configured as the primary root bridge of MSTI 1 and the secondary root bridge of MSTI 2. S2 is configured as the secondary root bridge of MSTI 1 and the primary root bridge of MSTI 2.
3. The IP address of each VLANIF interface is 10.0.x.y/24, where x indicates the VRID and y indicates the device ID. The virtual IP address is set to 10.0.x.254/24.

Initial Configuration

- Initial configuration of S1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S1
[S1]
```

- Initial configuration of S2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S2
[S1]
```

- Initial configuration of S3

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S3
[S3]
```

- Initial configuration of R1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]int lo0
[R1-LoopBack0]
[R1-LoopBack0]ip address 1.1.1.1 24
[R1-LoopBack0]quit
[R1]
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1]ip address 192.168.30.2 24
[R1-GigabitEthernet0/0/1]quit
[R1]
[R1]int g0/0/2
[R1-GigabitEthernet0/0/2]
[R1-GigabitEthernet0/0/2]ip address 192.168.40.2 24
[R1-GigabitEthernet0/0/2]quit
[R1]
```

- Network configuration of PC1

```
IP address: 10.0.10.10
Mask: 255.255.255.0
```

Gateway: 10.0.10.254

Task Implementation

1 Checking Basic Configurations

Omitted

2 Basic MSTP Configurations

Create VLANs 10 and 20 on all switches. Configure an MSTP region named **hcip**, and create two instances **Instance 1** and **Instance 2**. Map VLAN 10 to **Instance 1** and VLAN 20 to **Instance 2**. Plan S1 as the primary root bridge of MSTI 1 and secondary root bridge of MSTI 2; plan S2 as the primary root bridge of MSTI 2 and the secondary root bridge of MSTI 1.

VLAN 30 is used by S1 to connect to R1, and VLAN 40 is used by S2 to connect to R1.

1. Create VLANs.

```
[S1]vlan batch 10 20 30
```

```
Info: This operation may take a few seconds. Please wait for a moment...done.
```

```
[S1]
```

```
[S2]vlan batch 10 20 40
```

```
Info: This operation may take a few seconds. Please wait for a moment...done.
```

```
[S2]
```

```
[S3]vlan batch 10 20
```

```
Info: This operation may take a few seconds. Please wait for a moment...done.
```

```
[S3]
```

2. Configure all interconnection interfaces as trunk interfaces and allow packets from the corresponding VLANs to pass through.

– Configure S1.

```
[S1]int g0/0/10
```

```
[S1-GigabitEthernet0/0/10]port link-type trunk
```

```
[S1-GigabitEthernet0/0/10]port trunk allow-pass vlan 10 20
```

```
[S1-GigabitEthernet0/0/10]quit
```

```
[S1]
```

```
[S1]int g0/0/11
```

```
[S1-GigabitEthernet0/0/11]port link-type trunk
```

```
[S1-GigabitEthernet0/0/11]port trunk allow-pass vlan 10 20
```

```
[S1-GigabitEthernet0/0/11]quit
```

```
[S1]
```

```
[S1]int g0/0/12
```

```
[S1-GigabitEthernet0/0/12]port link-type trunk
```

```
[S1-GigabitEthernet0/0/12]port trunk allow-pass vlan 10 20
```

```
[S1-GigabitEthernet0/0/12]quit
```

```
[S1]
```

```
[S1]int g0/0/1
```

```
[S1-GigabitEthernet0/0/1]port link-type access
```

```
[S1-GigabitEthernet0/0/1]port default vlan 30
```

```
[S1-GigabitEthernet0/0/1]quit
```

```
[S1]
```

– Configure S2.

```
[S2]int g0/0/10
[S2-GigabitEthernet0/0/10]port link-type trunk
[S2-GigabitEthernet0/0/10]port trunk allow-pass vlan 10 20
[S2-GigabitEthernet0/0/10]quit
[S2]
[S2]int g0/0/11
[S2-GigabitEthernet0/0/11]port link-type trunk
[S2-GigabitEthernet0/0/11]port trunk allow-pass vlan 10 20
[S2-GigabitEthernet0/0/11]quit
[S2]
[S2]int g0/0/13
[S2-GigabitEthernet0/0/13]port link-type trunk
[S2-GigabitEthernet0/0/13]port trunk allow-pass vlan 10 20
[S2-GigabitEthernet0/0/13]quit
[S2]
[S2]int g0/0/2
[S2-GigabitEthernet0/0/2]port link-type access
[S2-GigabitEthernet0/0/2]port default vlan 40
[S2-GigabitEthernet0/0/2]quit
[S2]
```

– Configure S3.

```
[S3]int g0/0/1
[S3-GigabitEthernet0/0/1]port link-type trunk
[S3-GigabitEthernet0/0/1]port trunk allow-pass vlan 10 20
[S3-GigabitEthernet0/0/1]quit
[S3]
[S3]int g0/0/2
[S3-GigabitEthernet0/0/2]port link-type trunk
[S3-GigabitEthernet0/0/2]port trunk allow-pass vlan 10 20
[S3-GigabitEthernet0/0/2]quit
[S3]
[S3]int g0/0/10      # Interface for connecting to the PC.
[S3-GigabitEthernet0/0/10]port link-type access
[S3-GigabitEthernet0/0/10]port default vlan 10
[S3-GigabitEthernet0/0/10]quit
[S3]
```

3. Change the working mode from STP to MSTP.

```
[S1]stp mode mstp

[S2]stp mode mstp

[S3]stp mode mstp
```

4. Configure MSTP.

```
[S1]stp region-configuration
[S1-mst-region] region-name hcip
[S1-mst-region] revision-level 1
[S1-mst-region] instance 1 vlan 10
[S1-mst-region] instance 2 vlan 20
[S1-mst-region] active region-configuration
Info: This operation may take a few seconds. Please wait for a moment...done.
```

```
[S1-mst-region] quit

[S2]stp region-configuration
[S2-mst-region] region-name hcip
[S2-mst-region] revision-level 1
[S2-mst-region] instance 1 vlan 10
[S2-mst-region] instance 2 vlan 20
[S2-mst-region] active region-configuration
Info: This operation may take a few seconds. Please wait for a moment...done.
[S2-mst-region] quit

[S3]stp region-configuration
[S3-mst-region] region-name hcip
[S3-mst-region] revision-level 1
[S3-mst-region] instance 1 vlan 10
[S3-mst-region] instance 2 vlan 20
[S3-mst-region] active region-configuration
Info: This operation may take a few seconds. Please wait for a moment...done.
[S3-mst-region] quit
```

5. Check the mappings between MSTI instances and VLANs on S1.

```
[S1]display stp region-configuration
Oper configuration
Format selector      :0
Region name          :hcip
Revision level       :1

Instance  VLANs Mapped
  0         1 to 9, 11 to 19, 21 to 29, 31 to 39, 41 to 49, 51 to 59, 61 to
           69, 71 to 79, 81 to 4094
  1         10,
  2         20

[S1]
```

6. Configure S1 as the primary root bridge of MSTI 1 and the secondary root bridge of MSTI 2.

```
[S1]stp instance 1 root primary
[S1]stp instance 2 root secondary
```

7. Configure S2 as the primary root bridge of MSTI 2 and the secondary root bridge of MSTI 1.

```
[S2]stp instance 1 root secondary
[S2]stp instance 2 root primary
```

8. Check the status and statistics of MSTI 1 on S1.

```
[S1]display stp instance 1 brief
```

MSTID	Port	Role	STP State	Protection
1	GigabitEthernet0/0/10	DESI	FORWARDING	NONE
1	GigabitEthernet0/0/11	DESI	FORWARDING	NONE
1	GigabitEthernet0/0/12	DESI	FORWARDING	NONE

The command output shows that all interfaces on S1 are designated interfaces, and S1 is the root bridge of MSTI 1.

9. Check the status and statistics of MSTI 2 on S2.

```
[S2]display stp instance 2 brief
```

MSTID	Port	Role	STP State	Protection
2	GigabitEthernet0/0/10	DESI	FORWARDING	NONE
2	GigabitEthernet0/0/11	DESI	FORWARDING	NONE

2	GigabitEthernet0/0/13	DESI	FORWARDING	NONE
---	-----------------------	------	------------	------

The command output shows that all interfaces on S2 are designated interfaces, and S2 is the root bridge of MSTI 2.

3 Basic VRRP Configurations

Create VLANIF 10 and VLANIF 20 on both S1 and S2, and add VLANIF 10 to VRRP group 10 and VLANIF 20 to VRRP group 20. Configure VRRP priorities so that S1 in VLAN 10 and S2 in VLAN 20 both function as the VRRP master.

1. Create VLANIF interfaces.

```
[S1]interface Vlanif10
[S1-Vlanif10] ip address 10.0.10.1 255.255.255.0
[S1-Vlanif10] quit
[S1]interface Vlanif20
[S1-Vlanif20] ip address 10.0.20.1 255.255.255.0
[S1-Vlanif20] quit
[S1]
[S1]int Vlanif 30      #IP address of the interface connecting to R1
[S1-Vlanif30]ip address 192.168.30.1 24
[S1-Vlanif30]quit
[S1]

[S2]interface Vlanif10
[S2-Vlanif10] ip address 10.0.10.2 255.255.255.0
[S2-Vlanif10] quit
[S2]interface Vlanif20
[S2-Vlanif20] ip address 10.0.20.2 255.255.255.0
[S2-Vlanif20] quit
[S2]
[S2]int Vlanif 40      #IP address of the interface connecting to R1
[S2-Vlanif40]ip address 192.168.40.1 24
[S2-Vlanif40]quit
[S2]
```

2. Configure VRRP groups on S1.

```
[S1]interface Vlanif 10
[S1-Vlanif10] vrrp vrid 10 virtual-ip 10.0.10.254
[S1-Vlanif10] vrrp vrid 10 priority 120
[S1-Vlanif10] quit
[S1]interface Vlanif 20
[S1-Vlanif20] vrrp vrid 20 virtual-ip 10.0.20.254
[S1-Vlanif20] quit
```

Set the VRRP priority to 120 for S1 in VLAN 10, and use the default value 100 for S1 in VLAN 20.

3. Configure VRRP groups on S2.

```
[S2]interface Vlanif10
[S2-Vlanif10] vrrp vrid 10 virtual-ip 10.0.10.254
[S2-Vlanif10] quit
[S2]interface Vlanif20
[S2-Vlanif20] vrrp vrid 20 virtual-ip 10.0.20.254
[S2-Vlanif20] vrrp vrid 20 priority 120
[S2-Vlanif20] quit
```

Set the VRRP priority to 120 for S2 in VLAN 20, and use the default value 100 for S2 in VLAN 10.

4. Check the VRRP status.

```
[S1]display vrrp brief
VRID  State      Interface      Type      Virtual IP
-----
10    Master      Vlanif10      Normal    10.0.10.254
20    Backup     Vlanif20      Normal    10.0.20.254
-----
Total:2      Master:1      Backup:1      Non-active:0
[S1]

[S2]display vrrp brief
VRID  State      Interface      Type      Virtual IP
-----
10    Backup     Vlanif10      Normal    10.0.10.254
20    Master     Vlanif20      Normal    10.0.20.254
-----
Total:2      Master:1      Backup:1      Non-active:0
[S2]
```

The VRRP status is the same as expected.

4 Configuring Association Between VRRP and BFD to Implement Rapid Master/Backup VRRP Switchovers

Configure single-hop BFD on S1 and S2 to detect the connectivity between VLANIF interfaces. Associate VRRP with BFD so that the backup device increases its VRRP priority when the BFD session goes down.

1. Create BFD sessions on S1.

```
[S1]bfd
[S1-bfd] quit
[S1]bfd vlanif10 bind peer-ip 10.0.10.2 interface Vlanif10
[S1-bfd-session-vlanif10] discriminator local 1
[S1-bfd-session-vlanif10] discriminator remote 2
[S1-bfd-session-vlanif10] min-tx-interval 100
[S1-bfd-session-vlanif10] min-rx-interval 100
[S1-bfd-session-vlanif10] commit
[S1-bfd-session-vlanif10] quit
[S1]bfd vlanif20 bind peer-ip 10.0.20.2 interface Vlanif20
[S1-bfd-session-vlanif20] discriminator local 11
[S1-bfd-session-vlanif20] discriminator remote 22
[S1-bfd-session-vlanif20] min-tx-interval 100
[S1-bfd-session-vlanif20] min-rx-interval 100
[S1-bfd-session-vlanif20] commit
[S1-bfd-session-vlanif20] quit
```

2. Create BFD sessions on S2.

```
[S2]bfd
[S2-bfd] quit
[S2]bfd vlanif10 bind peer-ip 10.0.10.1 interface Vlanif10
[S2-bfd-session-vlanif10] discriminator local 2
[S2-bfd-session-vlanif10] discriminator remote 1
```



```
[S2-bfd-session-vlanif10] min-tx-interval 100
[S2-bfd-session-vlanif10] min-rx-interval 100
[S2-bfd-session-vlanif10] commit
[S2-bfd-session-vlanif10] quit
[S2]bfd vlanif20 bind peer-ip 10.0.20.1 interface Vlanif20
[S2-bfd-session-vlanif20] discriminator local 22
[S2-bfd-session-vlanif20] discriminator remote 11
[S2-bfd-session-vlanif20] min-tx-interval 100
[S2-bfd-session-vlanif20] min-rx-interval 100
[S2-bfd-session-vlanif20] commit
[S2-bfd-session-vlanif20] quit
```

3. Check the BFD session status.

```
[S1]display bfd session all
```

Local	Remote	PeerIpAddr	State	Type	InterfaceName
1	2	10.0.10.2	Up	S_IP_IF	Vlanif10
11	22	10.0.20.2	Up	S_IP_IF	Vlanif20
Total UP/DOWN Session Number : 2/0					

```
[S2]display bfd session all
```

Local	Remote	PeerIpAddr	State	Type	InterfaceName
2	1	10.0.10.1	Up	S_IP_IF	Vlanif10
22	11	10.0.20.1	Up	S_IP_IF	Vlanif20
Total UP/DOWN Session Number : 2/0					

The BFD sessions on S1 and S2 are in Up state.

4. Configure association between VRRP and BFD.

```
[S1]interface Vlanif20
[S1-Vlanif20] vrrp vrid 20 track bfd-session 11 increased 30
[S1-Vlanif20] quit

[S2]interface Vlanif10
[S2-Vlanif10] vrrp vrid 10 track bfd-session 2 increased 30
[S2-Vlanif10] quit
```

Note that *bfd-session-id* specifies the local discriminator of a BFD session. You only need to configure the backup device to track the BFD session. In this way, the backup device increases its local VRRP priority when the BFD session goes down.

5. Shut down all interfaces on S1 to simulate a link fault.

```
[S1]interface GigabitEthernet0/0/10
[S1-GigabitEthernet0/0/10] shutdown
[S1-GigabitEthernet0/0/10] quit
[S1]interface GigabitEthernet0/0/11
[S1-GigabitEthernet0/0/11] shutdown
[S1-GigabitEthernet0/0/11] quit
[S1]interface GigabitEthernet0/0/12
[S1-GigabitEthernet0/0/12] shutdown
[S1-GigabitEthernet0/0/12] quit
```

6. Check the BFD session status on S2.

```
<S2>display bfd session all
```

Local	Remote	PeerIpAddr	State	Type	InterfaceName
2	1	10.0.10.1	Down	S_IP_IF	Vlanif10
22	11	10.0.20.1	Down	S_IP_IF	Vlanif20
Total UP/DOWN Session Number : 0/2					

The two BFD sessions immediately enter the Down state.

7. Check the VRRP status on S2.

```
<S2>display vrrp brief
```

VRID	State	Interface	Type	Virtual IP
10	Master	Vlanif10	Normal	10.0.10.254
20	Master	Vlanif20	Normal	10.0.20.254
Total:2 Master:2 Backup:0 Non-active:0				

S2 functions as the master in both VRRP groups 10 and 20.

8. Check the VRRP status and configuration parameters on S2.

```
[S2]display vrrp
```

```
Vlanif10 | Virtual Router 10
State : Master
Virtual IP : 10.0.10.254
Master IP : 10.0.10.2
PriorityRun : 130
PriorityConfig : 100
MasterPriority : 130
Preempt : YES Delay Time : 0 s
TimerRun : 1 s
TimerConfig : 1 s
Auth type : NONE
Virtual MAC : 0000-5e00-010a
Check TTL : YES
Config type : normal-vrrp
Track BFD : 2 Priority increased : 30
BFD-session state : DOWN
Create time : 2020-06-05 11:01:54 UTC-08:00
Last change time : 2020-06-05 11:31:15 UTC-08:00
```

```
Vlanif20 | Virtual Router 20
State : Master
Virtual IP : 10.0.20.254
Master IP : 10.0.20.2
PriorityRun : 120
PriorityConfig : 120
MasterPriority : 120
Preempt : YES Delay Time : 0 s
TimerRun : 1 s
TimerConfig : 1 s
Auth type : NONE
Virtual MAC : 0000-5e00-0114
```

```
Check TTL : YES
Config type : normal-vrrp
Create time : 2020-06-05 11:01:54 UTC-08:00
Last change time : 2020-06-05 11:01:55 UTC-08:00
```

The priority of VRRP group 10 is 130, and the BFD session is in Down state. The BFD down event triggers S2 to increase the priority of VRRP group 10 by 30.

5 Configuring OSPF

1. Configure OSPF on R1, S1, and S2 to simulate an IP network.

```
[R1]
[R1]ospf 1 router-id 1.1.1.1
[R1-ospf-1]
[R1-ospf-1]area 0
[R1-ospf-1-area-0.0.0.0]
[R1-ospf-1-area-0.0.0.0]network 1.1.1.1 0.0.0.0
[R1-ospf-1-area-0.0.0.0]
[R1-ospf-1-area-0.0.0.0]network 192.168.30.2 0.0.0.0
[R1-ospf-1-area-0.0.0.0]
[R1-ospf-1-area-0.0.0.0]network 192.168.40.2 0.0.0.0
[R1-ospf-1-area-0.0.0.0]
[R1-ospf-1-area-0.0.0.0]quit
[R1-ospf-1]quit
[R1]
```

```
[S1]ospf 1
[S1-ospf-1]area 0
[S1-ospf-1-area-0.0.0.0]network 10.0.10.1 0.0.0.0
[S1-ospf-1-area-0.0.0.0]network 192.168.30.1 0.0.0.0
[S1-ospf-1-area-0.0.0.0]network 10.0.20.1 0.0.0.0
[S1-ospf-1-area-0.0.0.0]quit
[S1-ospf-1]quit
[S1]
```

```
[S2]ospf 1
[S2-ospf-1]area 0
[S2-ospf-1-area-0.0.0.0]network 10.0.10.2 0.0.0.0
[S2-ospf-1-area-0.0.0.0]network 192.168.40.1 0.0.0.0
[S2-ospf-1-area-0.0.0.0]network 10.0.20.2 0.0.0.0
[S2-ospf-1-area-0.0.0.0]quit
[S2-ospf-1]quit
[S2]
```

2. Check the OSPF neighbor and routing table information on R1.

```
<R1>display ospf peer brief
```

```
OSPF Process 1 with Router ID 1.1.1.1
Peer Statistic Information
```

Area Id	Interface	Neighbor id	State
0.0.0.0	GigabitEthernet0/0/2	10.0.10.2	Full
0.0.0.0	GigabitEthernet0/0/1	10.0.10.1	Full

```

-----
<R1>
<R1>display ospf routing

  OSPF Process 1 with Router ID 1.1.1.1
    Routing Tables

  Routing for Network
  Destination      Cost  Type      NextHop      AdvRouter      Area
  1.1.1.1/32       0     Stub      1.1.1.1      1.1.1.1        0.0.0.0
  192.168.30.0/24   1     Transit   192.168.10.2 1.1.1.1        0.0.0.0
  192.168.40.0/24   1     Transit   192.168.20.2 1.1.1.1        0.0.0.0
  10.0.10.0/24      2     Transit   192.168.10.1 10.0.10.1      0.0.0.0
  10.0.10.0/24      2     Transit   192.168.20.1 10.0.10.1      0.0.0.0
  10.0.20.0/24      2     Transit   192.168.10.1 10.0.10.1      0.0.0.0
  10.0.20.0/24      2     Transit   192.168.20.1 10.0.10.1      0.0.0.0

  Total Nets: 7
  Intra Area: 7  Inter Area: 0  ASE: 0  NSSA: 0

<R1>

```

Verification

1. On PC1, ping 1.1.1.1 of R1.

```

PC>ipconfig

Link local IPv6 address.....: fe80::5689:98ff:fe02:241e
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 10.0.10.10
Subnet mask.....: 255.255.255.0
Gateway.....: 10.0.10.254
Physical address.....: 54-89-98-02-24-1E
DNS server.....:

PC>ping 1.1.1.1 -c 3          # Check the network connectivity.

Ping 1.1.1.1: 32 data bytes, Press Ctrl_C to break
From 1.1.1.1: bytes=32 seq=1 ttl=254 time=172 ms
From 1.1.1.1: bytes=32 seq=2 ttl=254 time=47 ms
From 1.1.1.1: bytes=32 seq=3 ttl=254 time=47 ms

--- 1.1.1.1 ping statistics ---
  3 packet(s) transmitted
  3 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 47/88/172 ms

PC>tracert 1.1.1.1          # Check the data packet path.

tracert to 1.1.1.1, 8 hops max
(ICMP), press Ctrl+C to stop
 1  10.0.10.1  63 ms  47 ms  31 ms
 2  1.1.1.1   78 ms  63 ms  62 ms

```

```
PC>
```

2. Power off S1 and perform ping and tracer tests.

Continuously ping R1 from PC1 and add the **-t** parameter to the **ping** command.

```
PC>ping 1.1.1.1 -t
```

```
Ping 1.1.1.1: 32 data bytes, Press Ctrl_C to break
```

```
From 1.1.1.1: bytes=32 seq=1 ttl=254 time=93 ms
```

```
From 1.1.1.1: bytes=32 seq=2 ttl=254 time=63 ms
```

```
From 1.1.1.1: bytes=32 seq=3 ttl=254 time=62 ms
```

```
From 1.1.1.1: bytes=32 seq=4 ttl=254 time=47 ms
```

```
From 1.1.1.1: bytes=32 seq=5 ttl=254 time=63 ms
```

```
From 1.1.1.1: bytes=32 seq=6 ttl=254 time=47 ms
```

```
From 1.1.1.1: bytes=32 seq=7 ttl=254 time=47 ms
```

```
From 1.1.1.1: bytes=32 seq=8 ttl=254 time=78 ms
```

```
From 1.1.1.1: bytes=32 seq=9 ttl=254 time=47 ms
```

Power off S1 and observe the connectivity and path of data packets for about 60 seconds.

```
PC>ping 1.1.1.1 -t
```

```
Ping 1.1.1.1: 32 data bytes, Press Ctrl_C to break
```

```
From 1.1.1.1: bytes=32 seq=1 ttl=254 time=93 ms
```

```
From 1.1.1.1: bytes=32 seq=2 ttl=254 time=63 ms
```

```
From 1.1.1.1: bytes=32 seq=3 ttl=254 time=62 ms
```

```
From 1.1.1.1: bytes=32 seq=4 ttl=254 time=47 ms
```

```
From 1.1.1.1: bytes=32 seq=5 ttl=254 time=63 ms
```

```
From 1.1.1.1: bytes=32 seq=6 ttl=254 time=47 ms
```

```
From 1.1.1.1: bytes=32 seq=7 ttl=254 time=47 ms
```

```
From 1.1.1.1: bytes=32 seq=8 ttl=254 time=78 ms
```

```
From 1.1.1.1: bytes=32 seq=9 ttl=254 time=47 ms
```

```
From 1.1.1.1: bytes=32 seq=10 ttl=254 time=46 ms
```

```
From 1.1.1.1: bytes=32 seq=11 ttl=254 time=63 ms
```

```
Request timeout!
```

```
Request timeout!
```

```
Request timeout!
```

```
Request timeout!
```

```
Request timeout!
```

```
Request timeout!
```

```
From 1.1.1.1: bytes=32 seq=30 ttl=254 time=125 ms
```

```
From 1.1.1.1: bytes=32 seq=31 ttl=254 time=47 ms
```

```
From 1.1.1.1: bytes=32 seq=32 ttl=254 time=78 ms
```

```
From 1.1.1.1: bytes=32 seq=33 ttl=254 time=47 ms
```

```
From 1.1.1.1: bytes=32 seq=34 ttl=254 time=63 ms
```

```
From 1.1.1.1: bytes=32 seq=35 ttl=254 time=46 ms
```

```
From 1.1.1.1: bytes=32 seq=36 ttl=254 time=47 ms
```

```
From 1.1.1.1: bytes=32 seq=37 ttl=254 time=47 ms
```

```
From 1.1.1.1: bytes=32 seq=38 ttl=254 time=47 ms
```

```
From 1.1.1.1: bytes=32 seq=39 ttl=254 time=93 ms
```

Run the tracer command, with S1 powered off.

```
PC>tracert 1.1.1.1
```

```
tracert to 1.1.1.1, 8 hops max
```

```
(ICMP), press Ctrl+C to stop
```

```
1 10.0.10.2 47 ms 32 ms 46 ms
2 1.1.1.1 79 ms 62 ms 78 ms
```

PC>

The data packet path shows that the data packet has been forwarded by S2.

Quiz

1. How can I adjust the interval between gratuitous ARP packets sent from a VRRP group?

By default, the master router in a VRRP group sends a gratuitous ARP packet every 2 minutes. You can use the **vrrp gratuitous-arp timeout** command to change the interval at which gratuitous ARP packets are sent.

2. What are the possible causes for two master devices in a VRRP group?

The possible causes are as follows:

- Configuration parameters, such as the authentication mode and key, VRRP group ID, virtual address list, and version, on devices in the VRRP group are different.
- The link where heartbeat packets pass is unstable or disconnected.
- The interface is blocked by loop prevention protocols such as STP and RRPP.
- The CPU usage is high.

3. How is the VRRP status negotiated when the same priority is set for devices in the VRRP group?

If the VRRP master and backup are determined, the VRRP status is not negotiated again. If two masters exist before devices in the VRRP group exchange heartbeat packets, the VRRP status is negotiated as follows:

If the source IP address of a heartbeat packet received by the local end is larger than the primary IP address of the interface, the local device switches to the Backup state. Otherwise, the local end discards the heartbeat packet and remains in the Master state. The peer end switches to the Backup state when the local end sends a heartbeat packet with a source IP address larger than the IP address of the peer end.